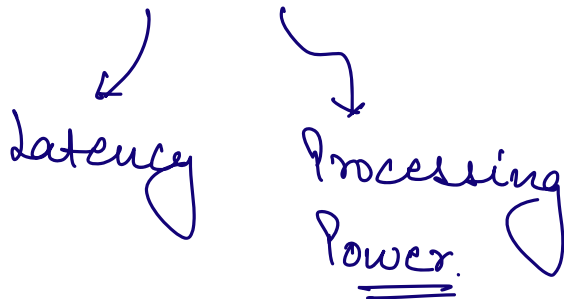


Agenda.

- Prompt Engineering
- Prompting techniques.
- Prompting Parameters.
- Optimize Prompts.

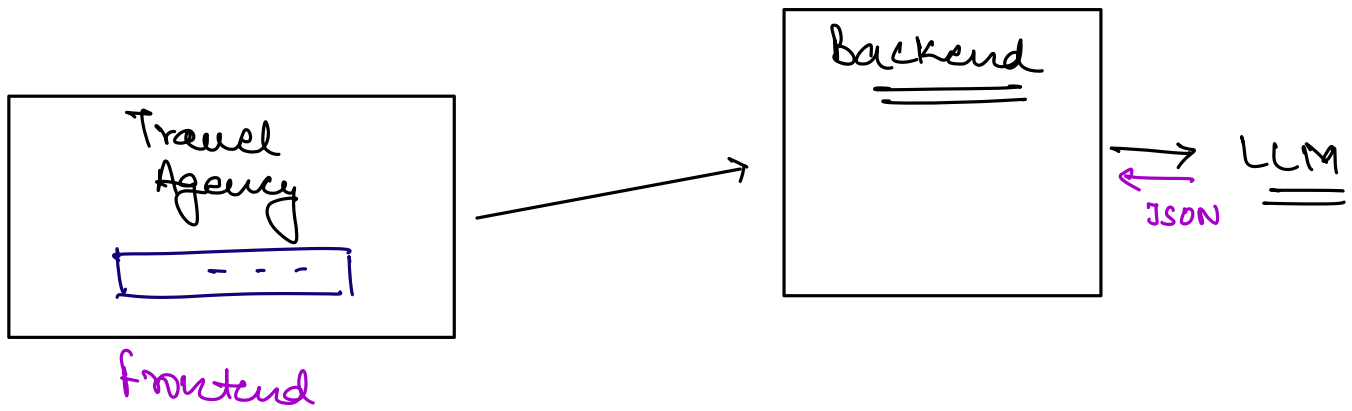
Prompt Engineering is an art of writing instructions to guide the LLM for better outputs & performance.



elements of a prompt

(task)

- Instruction → What we want LLM to do?
- Context → Additional info | background of the task.
- Constraints → limitations on the response
- Format → Response structure.
- Variables → Template {----} (JSON format)



Prompt Parameters.

→ To control the LLM's output & performance.

1. Temperature

2. Sampling

3. Repetition Penalty

4. Max tokens.

Temperature.

↳ Controls the randomness / creativity.
of model.

⇒ Range: 0-2

Low temperature $\Rightarrow$ More precise | deterministic |
predictable output

High temperature $\Rightarrow$ More Creativity | randomness |
diverse.

Temperature

0 - 0.5

Example | Use-cases

Algorithms | Code | Financial |
Maths | analysis | . . .

0.5 - 1

Posts | Blog | Chat App | . . .

1 - 1.5

Story telling | Fictional |
Novels.

Highly
Creative

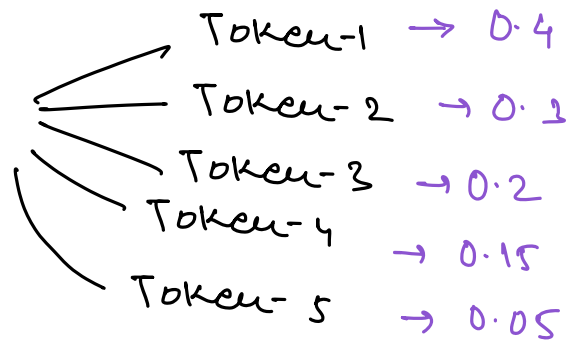
1.5 - 2

$\rightarrow$ X

Sampling

LLM

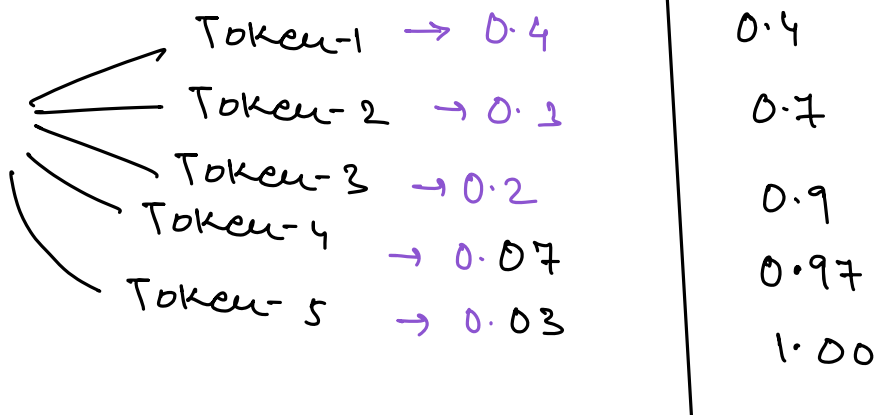
↳ Controls how the model picks up the tokens based on the probability.



Top-k

Top-p

↳ P = 0.9



Top-p → 0.7

T-1 → 0.25
T-2 → 0.22
T-3 → 0.18
T-4 → 0.15
T-5 → 0.12
T-6 → 0.8

0.25 ↓
0.47
0.65 →
0.80
0.92
1.00

Repetition Penalty.

↓

Controls the repetition of words or tokens in the response generated by LLM.

→ helps us to reduce redundancy.

1 → No penalty
↳ Repitition is allowed

1-1.5 → mild repitition is allowed

1.5-2 → Strong control on repitition
(No repitition)

>2 → ✗

Max Tokens

→ Maximum no. of tokens LLM can generate
in the response.

Prompting Techniques

- Zero Shot Prompting
- One Shot Prompting
- few Shot Prompting
- Role Prompting
- Chain-Of-Thought
- Self Consistency.

Zero Shot Prompting

↳ Just give the task to LLM, without any example.

One Shot Prompting

give the task to LLM, with exactly one example.

Few Shot Prompting

give the task to LLM, with (n) no. of examples

Role Prompting

↳ Assign a role to the model inside the prompt.

Chain-Of-Thought

↳ Show reasoning step-by-step
↳ Ask the model to give the response in a step-by-step manner with reasoning.

→ Instruction + Step-by-step reasoning.

Ex:

Maths problem

Coding / Algorithmic problem.

Self-Consistency

→ Instead of trusting one answer, self-consistency prompt asks the model:

→ Generate multiple answers.

→ Compare these answers.

→ Select the most frequent / Consistent answer.

→ Use Case: Where more predictable outputs are expected.

Optimal Prompt for Agents.

→ Add more context

→ Make the prompt clear

→ Add some examples

→ Add steps.

→ Add security / guardrailing